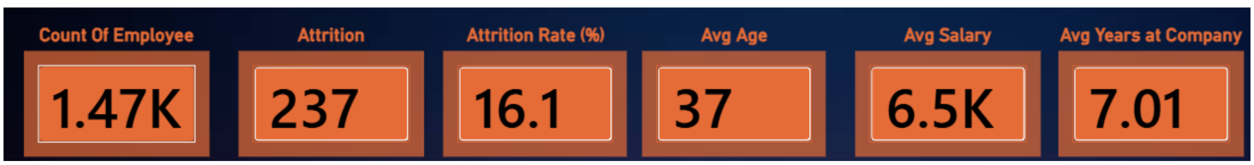
Employee Attrition Analysis

Report - EDA, Data Preprocessing & Power BI

1. Project Snapshot

Analyze 1,470 employee records with 35 HR features to identify patterns associated with employee attrition and present actionable HR segments through Power BI.

Item	Value
Records	1,470 employees
Features	35 HR features
Attrition cases	237
Overall attrition rate	16.1%
Target	Attrition (Yes/No)
Tools	Pandas, NumPy, Matplotlib, Seaborn, Power BI



Dashboard KPI: workforce = 1.47K, attrition = 237, attrition rate = 16.1%, average age = 37, average salary = 6.5K, average tenure = 7.01 years.

2. Data Cleaning & Preprocessing

- Inspected dataset shape, sample rows, data types and column structure.
- Checked missing values, duplicate rows, constant columns, identifier-like columns and unexpected data types.
- Removed exact duplicate rows.
- Standardized categorical text by converting object columns to strings and trimming whitespace.
- Removed constant columns: EmployeeCount, Over18 and StandardHours.
- Kept EmployeeNumber for traceability but excluded it from analytical/correlation features because it is an identifier.
- Created AttritionFlag: Yes = 1, No = 0 for numerical analysis and correlation.

3. Baseline Attrition

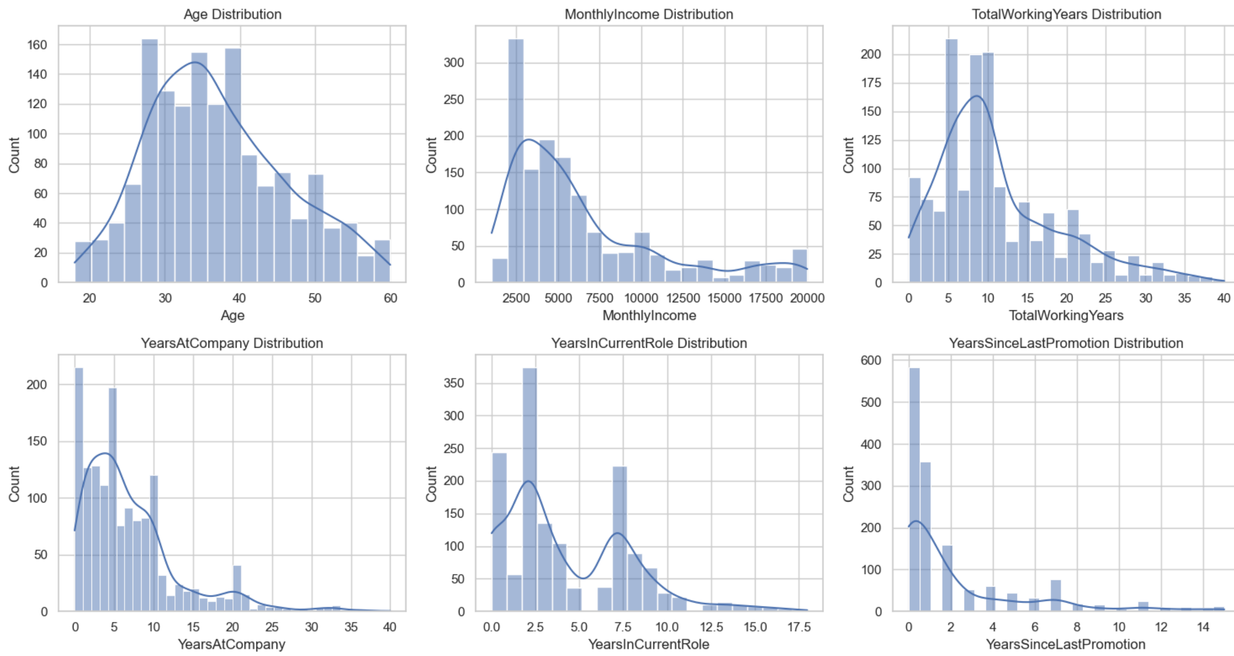
237 of 1,470 employees experienced attrition, giving an overall attrition rate of approximately 16.1%.

This is the baseline against which the later segments are compared.

4. Univariate EDA

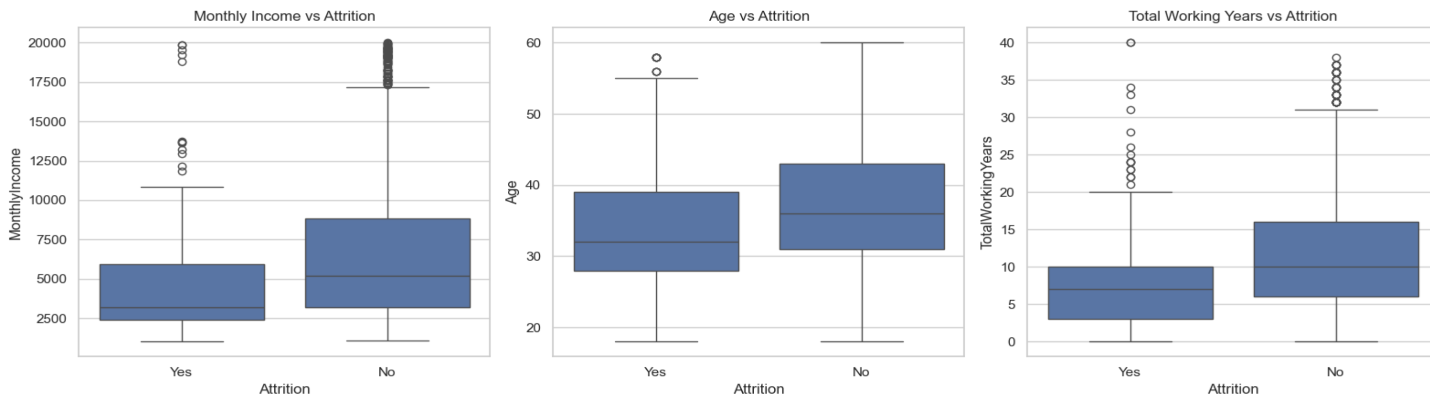
Key numerical features were summarized using count, mean, standard deviation, quartiles, minimum and maximum. The main distributions were then visualized.

	count	mean	std	min	25%	50%	75%	max
Age	1473.0	36.92	9.13	18.0	30.0	36.0	43.0	60.0
MonthlyIncome	1473.0	6500.23	4706.05	1009.0	2911.0	4908.0	8380.0	19999.0
JobLevel	1473.0	2.06	1.11	1.0	1.0	2.0	3.0	5.0
TotalWorkingYears	1473.0	11.28	7.78	0.0	6.0	10.0	15.0	40.0
YearsAtCompany	1473.0	7.00	6.12	0.0	3.0	5.0	9.0	40.0
YearsInCurrentRole	1473.0	4.23	3.62	0.0	2.0	3.0	7.0	18.0
YearsSinceLastPromotion	1473.0	2.18	3.22	0.0	0.0	1.0	3.0	15.0



Feature	Key result	Takeaway
Age	Mean 36.92; median 36; range 18–60	Workforce mainly concentrated around 25–45.
MonthlyIncome	Mean 6,500.23; median 4,908	Right-skewed; higher salaries pull the mean upward.
JobLevel	Mean 2.06; median 2; range 1–5	Typical employee is around Level 2.
TotalWorkingYears	Mean 11.28; median 10; range 0–40	Experience concentrated around 5–15 years.
YearsAtCompany	Mean 7; median 5; range 0–40	Mostly short-to-moderate tenure.
YearsInCurrentRole	Mean 4.23; median 3; range 0–18	Most employees have relatively short current-role tenure.
YearsSinceLastPromotion	Mean 2.18; median 1; range 0–15	Most are within a few years of last promotion.

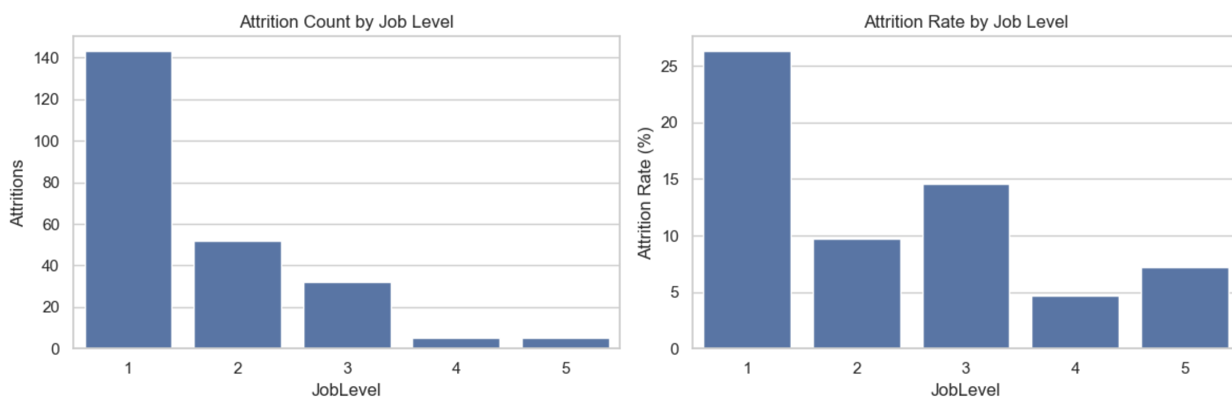
5. Bivariate Analysis — Attrition vs Income, Age & Experience



Variable	Observed pattern
Monthly Income	Median is ≈3.2K for employees who left vs ≈5.2K for those who stayed.
Age	Median is ≈32 for employees who left vs ≈36 for those who stayed.
Total Working Years	Median is ≈7 years for employees who left vs ≈10 for those who stayed.

Employees who left generally had lower income, were younger, and had fewer total working years.

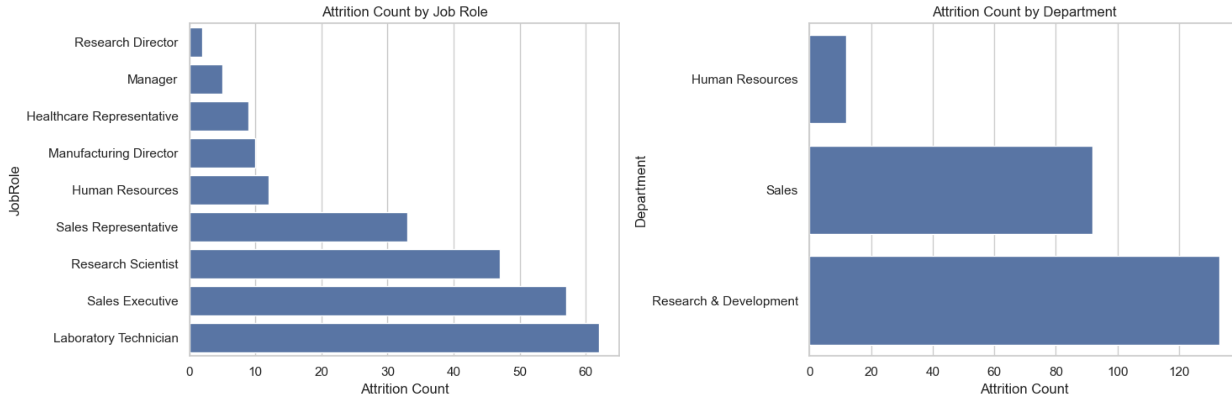
6. Attrition by Job Level



Job Level	Attrition Rate	Interpretation
1	~26%	Highest
3	~14.5%	Second-highest
2	~9.7%	Moderate
5	~7.0%	Lower
4	~4.5%	Lowest

Count tells how many employees left; rate tells what percentage of that job-level population left. Rate is the fairer comparison when group sizes differ.

7. High-Attrition Roles & Departments

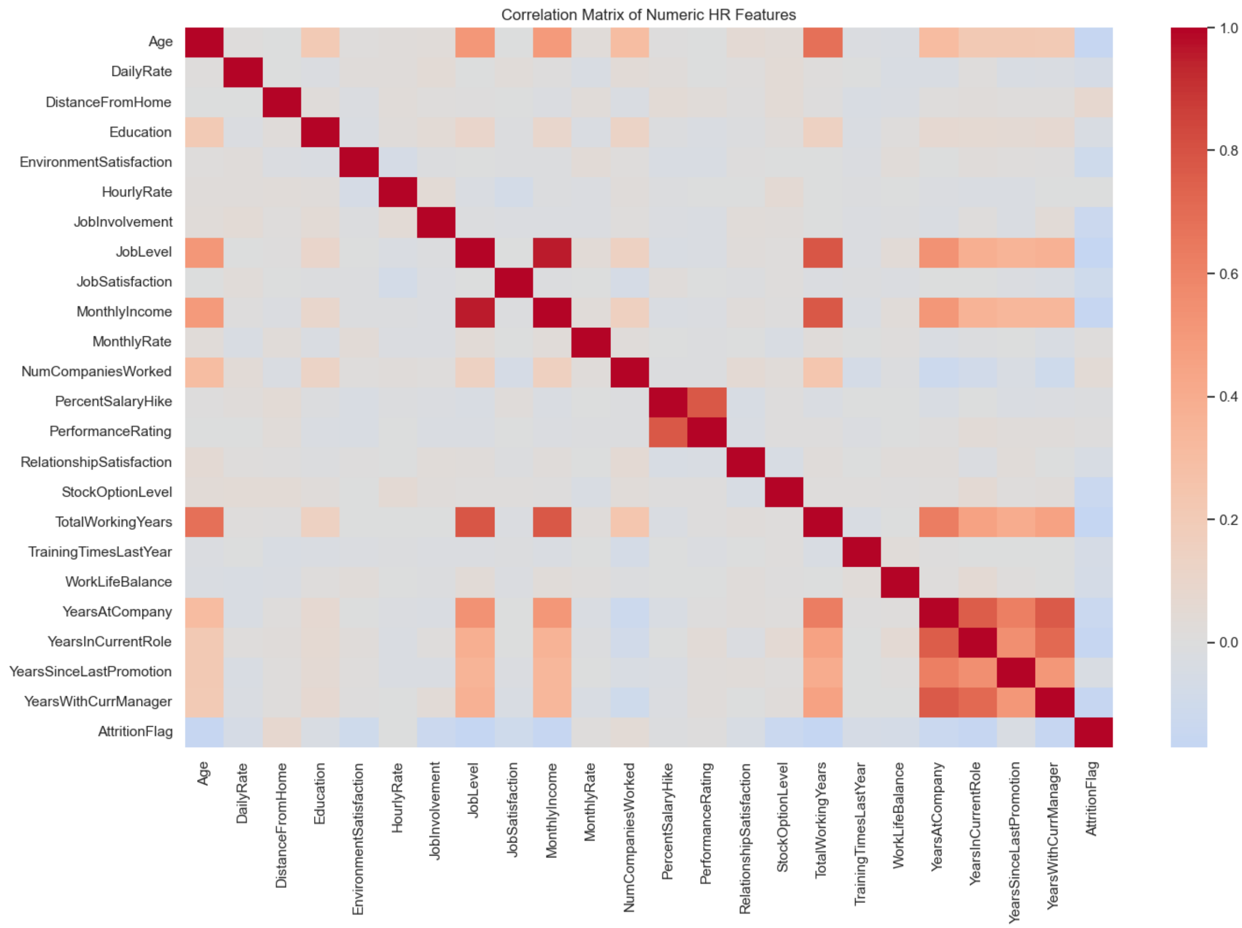


Job Role	Attrition Count
Laboratory Technician	62
Sales Executive	57
Research Scientist	47
Sales Representative	33

Department	Attrition Count
Research & Development	~133
Sales	~91
Human Resources	~12

These counts identify segments for deeper investigation. For unequal group sizes, compare attrition rates as well as counts.

8. Correlation Analysis



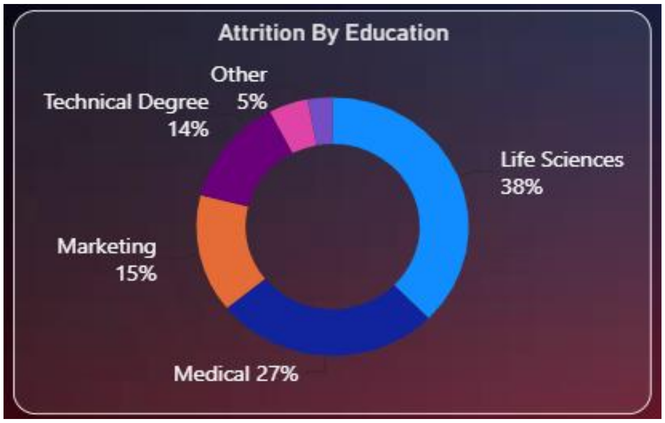
Correlation measures the strength and direction of a linear relationship from -1 to +1. The heatmap shows relationships among numerical HR features; the direct AttritionFlag correlations identify which numerical features are most associated with attrition.

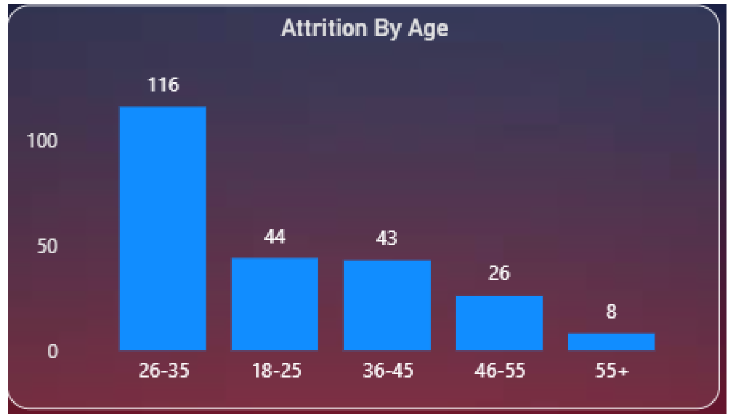
	Correlation with Attrition
TotalWorkingYears	-0.170847
JobLevel	-0.168926
YearsWithCurrManager	-0.161636
YearsInCurrentRole	-0.160302
MonthlyIncome	-0.159458
Age	-0.158775
StockOptionLevel	-0.136939
YearsAtCompany	-0.134106
JobInvolvement	-0.129678
EnvironmentSatisfaction	-0.104022
JobSatisfaction	-0.103276
DistanceFromHome	0.077585
WorkLifeBalance	-0.064221
TrainingTimesLastYear	-0.059769
DailyRate	-0.056809
RelationshipSatisfaction	-0.045763
NumCompaniesWorked	0.043469
YearsSinceLastPromotion	-0.032487
Education	-0.030526
MonthlyRate	0.014647
PercentSalaryHike	-0.013827
HourlyRate	-0.005593
PerformanceRating	0.003268

Feature	Correlation with Attrition
TotalWorkingYears	-0.171
JobLevel	-0.169
YearsWithCurrManager	-0.162
YearsInCurrentRole	-0.160
MonthlyIncome	-0.159
Age	-0.159
StockOptionLevel	-0.137
YearsAtCompany	-0.134
JobInvolvement	-0.130

Interpretation: the strongest associations are negative but weak. Higher experience, job level, income and age are associated with lower AttritionFlag values in this dataset. This does not establish causation.

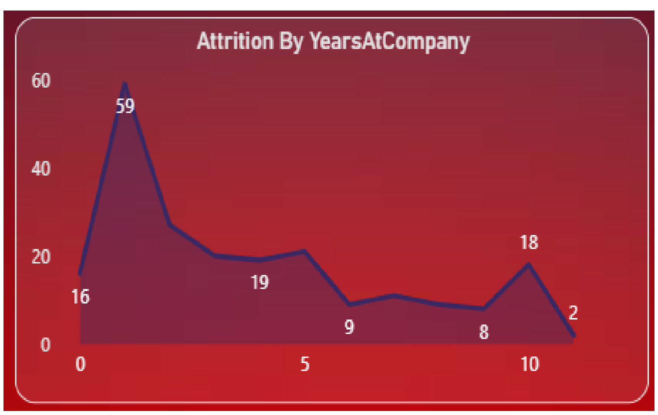
9. Power BI Dashboard

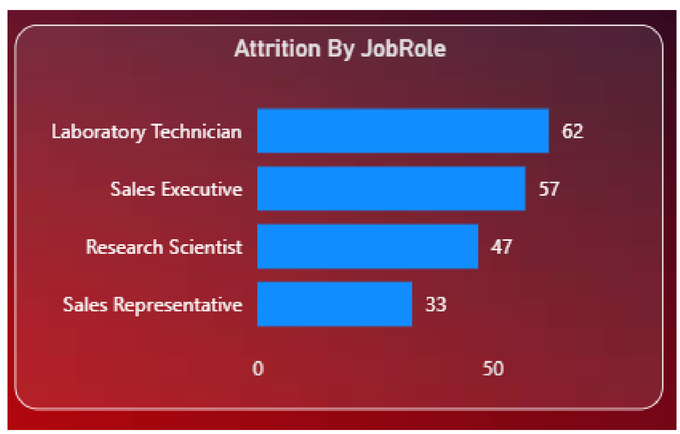




Job Satisfaction By Roles

JobRole	1	2	3	4	Total
Healthcare Representative	2	2	1	4	**9**
Human Resources	5	2	3	2	**12**
Laboratory Technician	20	8	21	13	**62**
Manager	1	2	1	1	**5**
Manufacturing Director	2	2	4	2	**10**
Research Director	0	1	1	0	**2**
Research Scientist	13	10	15	9	**47**
Sales Executive	16	9	18	14	**57**
Sales Representative	7	10	9	7	**33**
Total	**66**	**46**	**73**	**52**	**237**





Selected dashboard visuals show how the EDA findings were communicated: age groups, salary slabs, high-attrition roles, education, job satisfaction, tenure and gender. The dashboard uses different visual forms to answer different questions: KPI cards for headline metrics, bars for ranking, a matrix for role/satisfaction detail, a donut for composition, and a line chart for tenure pattern.

10. Final Business Insights

- Overall attrition is 16.1%: 237 of 1,470 employees left.
- Job Level 1 has the highest attrition rate at about 26%, while Level 4 is lowest at about 4.5%.
- Employees who left generally had lower monthly income, younger age and fewer total working years.
- The strongest numerical associations with AttritionFlag are TotalWorkingYears (-0.171), JobLevel (-0.169), YearsWithCurrManager (-0.162), YearsInCurrentRole (-0.160), MonthlyIncome (-0.159) and Age (-0.159).
- Highest attrition counts occur in Laboratory Technician, Sales Executive, Research Scientist and Sales Representative roles.
- Research & Development has the highest department attrition count, followed by Sales.
- These findings identify patterns and segments for further HR investigation; they do not prove causation.

The key takeaway was that attrition was more associated with younger, lower-income and lower-level employees, with shorter experience or tenure also showing negative associations. I segmented attrition by job level, role and department and used Power BI to make these patterns accessible for HR decision-making.